Yushan Wei

yushanwe@umich.edu | (734) 882-1215 | Ann Arbor, MI

Education

University of Michigan, Ann Arbor Sep. 2025 – Apr. 2027 (Expected)

B.S.E. in Computer Science

GPA: 3.94/4.00

Shanghai Jiao Tong University Sep. 2023 – Aug. 2027 (Expected)

B.S.E. in Mechanical Engineering (Dual Degree Program)

John Wu & Jane Sun Sunshine Scholarship (2024); University Physics Competition Silver Medal (2023)

Research Experience

SceneCopilot: Human-AI Co-Creation of 3D Scenes by Integrating Editable Zones and Constraints Oct. 2025 – Apr. 2026

Yushan Wei, Abhijeet Sahdev, Lei Zhang · Submitted to ACM CHI 2027

- Developed a reference-guided scene planning pipeline that retrieves and ranks floor plans, enables user selection, and extracts semantic scene sections using LLMs.
- Designed and implemented an editable section-based representation for 3D scenes, supporting interactive layout editing, spatial constraints, and automatic wall, door, and window generation.
- Extended a hierarchical constraint graph to link sections with their contained objects, enabling scene and section resizing while preserving spatial relationships through layout adjustments.

ATMaker: Conversationally Authoring Visual Assistive Tools May 2026 – Sep. 2026

Ellie Seehorn, **Yushan Wei**, Venkatesh Potluri, Anhong Guo · Submitted to ACM CHI 2027

- Extended an accessible system enabling blind and low-vision users to create bespoke camera-based visual assistive tools through natural language and coding agents.
- Designed LLM-guided routing strategies to select vision models and execution policies across visual tasks, balancing response accuracy and latency.
- Restructured the tool-generation pipeline around reusable agent abstractions, enabling coding agents to compose tools through supported interfaces while facilitating integration with wearable devices including Meta Ray-Ban smart glasses.
- Contributed to the design, execution, and analysis of user and researcher studies evaluating system usability and how effectively the system captures users' intent.

Driving Simulator for Human Factors Research Jan. 2026 – Dec. 2026

Subteam Leader · University of Michigan Transportation Research Institute

- Developed AI traffic vehicles in Unreal Engine with path following, obstacle avoidance, and location-triggered behaviors such as stopping and overtaking.
- Modeled configurable traffic scenarios using driving data, enabling precise control over surrounding vehicles' positions, speeds, following distances, and acceleration profiles.
- Built a data-driven scene transition system that seamlessly switches traffic conditions during visual occlusion, enabling rapid prototyping and testing of multiple experimental scenarios.

Teaching Experience

EECS 445 Introduction to Machine Learning Sep. 2026 – Dec. 2026

Instructional Aide · University of Michigan

- Led discussion sessions and held office hours, helping students understand machine learning concepts and solve technical problems.
- Collaborated with the teaching team to develop, review, and refine homework assignments and projects for technical correctness and clarity.

Selected Course Projects

EECS 482 Introduction to Operating Systems Jan. 2026 – Apr. 2026

Team Lead · University of Michigan

- Implemented a multi-threaded delivery system using kernel-level synchronization primitives, including mutexes, condition variables, context switching, and interrupt handling.
- Built a virtual memory pager managing process address spaces, page tables, physical memory, and disk-backed storage through page faults and system calls.

Industry Experience

Alcatel-Lucent Enterprise Dec. 2023 – Jan. 2024

Marketing Intern · Shanghai, China

- Developed JavaScript-based features for the company's web portal and proposed and implemented three structural improvements.

Leadership and Activities

Student Science and Technology Innovation Association Apr. 2024 – Aug. 2025

Communications Department Head · Shanghai Jiao Tong University

- Led science outreach initiatives including competitions, workshops, and faculty lab tours; secured over **20,000 RMB** in sponsorship.
- Organized machine learning workshops, mentoring student teams on applied ML projects.
- Managed official blog and media channels to connect undergraduates with research opportunities.

College Debate Team Sep. 2024 – Aug. 2025

Team Captain · Shanghai Jiao Tong University

- Represented the school at International Chinese Debating Competition, achieving **Top 8 finish** as a team.
- Led the college debate team in university-wide competitions, served as a tournament judge, and conducted regular training sessions, including mock debates and debate analysis.
- Organized exhibition matches, practice tournaments, and recruitment events to engage and develop new members.

Technical Skills

Programming: Python, C++, C, JavaScript, Swift, MATLAB, HTML/CSS

AI & ML: PyTorch, NumPy, pandas, OpenCV, LLM/VLM integration

Frameworks & Tools: React Native, SwiftUI, FastAPI, Unreal Engine, Git, Linux